Assignment 2

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Student Academic Performance Dataset

Task 1: Data Quality Issues

The dataset was examined to identify data quality issues. Missing values were found in Study_Hours, Attendance(%), and Grade. In addition, the dataset contained duplicate records that could affect the analysis.

```
Missing Values:
Student_ID       0
Age              0
Gender           0
Study_Hours     31
Attendance(%)   30
Test_Score       0
Grade            9
dtype: int64

Duplicates:
20

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1020 entries, 0 to 1019
Data columns (total 7 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Student_ID      1020 non-null   object
 1   Age             1020 non-null   int64
 2   Gender          1020 non-null   object
 3   Study_Hours     989 non-null    float64
 4   Attendance(%)   990 non-null    float64
 5   Test_Score      1020 non-null   float64
 6   Grade           1011 non-null   object
dtypes: float64(3), int64(1), object(3)
memory usage: 55.9+ KB
```

Task 2: Missing Value Handling

Missing values in Study_Hours and Attendance(%) were replaced using the median because it is less affected by outliers. Missing values in Grade were replaced using the mode because it is a categorical feature.

```
Student_ID       0
Age              0
Gender           0
Study_Hours      0
Attendance(%)    0
Test_Score       0
Grade            0
dtype: int64
```

Task 3: Outlier Detection and Handling Using IQR

Outliers were detected and removed using the Interquartile Range (IQR) method. After removing outliers, the dataset size was reduced from 1020 rows to 994 rows.

```
print(df.shape)

Dataset Shape After Removing Outliers:
(994, 7)
```

Task 4: Feature Normalization

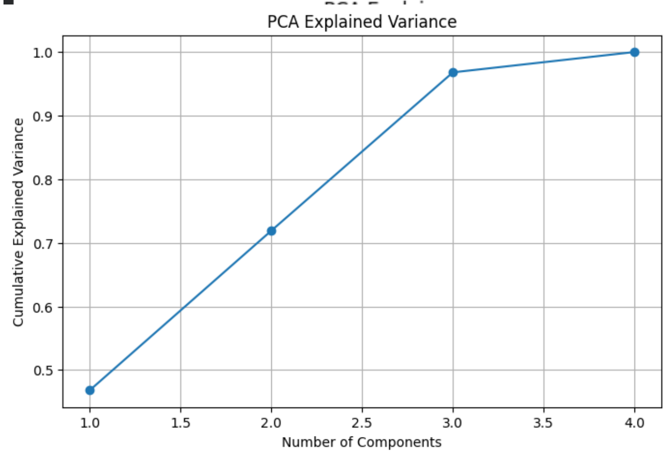
Numerical features were normalized using Min-Max Scaling and Z-Score Standardization. These techniques ensure that numerical features are on comparable scales and improve machine learning performance.

```
Min-Max Scaling
        Age  Study_Hours  Attendance(%)  Test_Score
0  0.500000     0.272727       0.292220        0.37
1  0.166667     0.151515       0.654649        0.26
2  0.333333     0.686869       0.732448        0.88
3  0.000000     0.626263       0.662239        0.55
4  0.500000     0.747475       0.770398        0.77

Z-Score Scaling
        Age  Study_Hours  Attendance(%)  Test_Score
0  0.008016    -1.247736      -1.143716   -0.619098
1 -0.988016    -1.897251       0.877066   -1.149311
2 -0.490000     0.971441       1.310846    1.839159
3 -1.486032     0.646684       0.919386    0.248522
4  0.008016     1.296199       1.522447    1.308947
```

Task 5: Principal Component Analysis (PCA)

PCA was applied to analyze explained variance and reduce dimensionality. The first principal component explained approximately 46.83% of the variance, while the first three principal components explained about 96.8% of the total variance.

```
Explained Variance Ratio:
[0.46829902 0.25086838 0.24887932 0.03195329]

Cumulative Explained Variance:
[0.46829902 0.71916739 0.96804671 1.        ]
```



Conclusion

The dataset was successfully preprocessed through data quality assessment, missing value handling, outlier removal, feature normalization, and dimensionality reduction using PCA. These preprocessing steps improved data quality and prepared the dataset for future machine learning applications.